

OPTICS B: ANSWER KEY

Section 1. Multiple Choice. 1 point each.

1. B
2. C
3. B
4. C
5. A
6. A
7. C
8. A
9. B
10. B
11. A
12. B
13. D
14. D
15. A
16. C
17. B
18. C
19. B
20. B
21. D

Section 2. Short answer. 3 points possible for each, but partial credit may be awarded for partially correct/complete answers.

22. **(TIE BREAKER)** MUST MENTION: The light should internally reflect through the fiber.
23. $\arcsin(1/1.9) = 32.07$
24. 19.47 deg
25. MUST MENTION: Printers use subtractive color model. They use black, magenta, cyan, and yellow.

Section 3. Long answer. 5 points possible each. Partial credit may be awarded for partially correct/complete answers. Deduct points if information not formatted in full sentences.

26. Accept definitions of Snell's law that include some variation of:
a formula used to describe the relationship between the angles of incidence and refraction, when referring to light passing through a boundary between two different media

Must include at least $\sin(\theta_1)/\sin(\theta_2)=n_2/n_1$

$$\frac{\sin \theta_1}{\sin \theta_2} = n_{21} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

27. Answers must include some variation of the following:

- a. Real images are from a converging lens or concave mirror, while virtual images are produced using diverging lens and convex mirrors when rays appear to converge but do not actually do so physically.
- b. Real images are inverted while virtual images are upright/erect
- c. Real images are on the screen, but virtual images cannot be projected.

26. **(TIE BREAKER)** Students may select to define any three of the following:

Anterior Chamber - Space between the iris and cornea that is filled with aqueous humor

Choroid - Supplies the retina with nutrients, and maintains eye temperature

Cones- Color receptors (optional: for daytime)

Lacrimal Caruncle - Contributes to eye lubrication and maintenance of the tear film

Lateral Rectus Muscle - Responsible for abduction

Optic Nerve - Transmits visual information from the retina to the brain

Vitreous Body - Fills the space between the retina and lens of the eye